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CMPT 477
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September 29, 2026

Project Three

2. **Source Image:** Lena_Y.tif

Filtering: Differential Rank Impulse Detector

Clean Image	Noisy Image (random w/ $p=0.05$)	Differential Rank Filtering (3x3 filter, $r=2$, $s=10$)
	 RMSE: 19.641, PSNR: 22.268	 RMSE: 3.716, PSNR: 36.729

Clean Image	Noisy Image (random w/ $p=0.1$)	Differential Rank Filtering (3x3 filter, $r=2$, $s=10$)
	 RMSE: 27.231, PSNR: 19.43	 RMSE: 6.402, PSNR: 32.004

Clean Image	Noisy Image (salt and pepper w/ $p=0.05$)	Differential Rank Filtering (3x3 filter, $r=2$, $s=10$)
	 RMSE: 29.948, PSNR: 18.604	 RMSE: 9.183, PSNR: 28.871

Second Differential Rank Pass (3x3 filter, $r=2$, $s=10$)	Third Differential Rank Pass (3x3 filter, $r=2$, $s=10$)	Fourth Differential Rank Pass (3x3 filter, $r=3$, $s=10$)
 RMSE: 4.2413, PSNR: 35.581	 RMSE: 3.667, PSNR: 36.845	 RMSE: 3.473, PSNR: 37.318

Fifth Differential Rank Pass (3x3 filter, r=3, s=20)	Sixth Differential Rank Pass (3x3 filter, r=2, s=40)
 RMSE: 3.073, PSNR: 38.381	 RMSE: 2.778, PSNR: 39.257

Clean Image	Noisy Image (salt and pepper w/ p=0.1)	Differential Rank Filtering (3x3 filter, r=2, s=10)
	 RMSE: 42.138, PSNR: 15.637	 RMSE: 18.453, PSNR: 22.809

Second Differential Rank Pass (3x3 filter, r=3, s=10)	Third Differential Rank Pass (3x3 filter, r=3, s=5)	Fourth Differential Rank Pass (3x3 filter, r=3, s=10)
 RMSE: 9.494, PSNR: 28.582	 RMSE: 6.642, PSNR: 31.685	 RMSE: 5.075, PSNR: 34.021

Fifth Differential Rank Pass(3x3 filter, r=2, s=40)
 RMSE: 4.716, PSNR: 34.66

Conclusion:

Differential Rank Impulse Detection is a very effective method for removing impulse noise from an image even with a high percentage of affected pixels, such as 10%. When cleaning an image affected by random impulse noise, a single iteration was able to remove most of the noise and significantly increase the PSNR. When cleaning an image affected by salt and pepper noise, several iterations were needed to remove all of the noise and increase the PSNR.